

HCSM-16

Loop Operator on the Full 56-Dimensional Hodge Sector

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Abstract

The Hodge Complex Standard Model derives the gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$ (HCSM-06), and determines which part of that algebra commutes with the loop operator M_{loop} and with the DNLS Hartree Hamiltonian H_{DNLS} on the 14-dimensional visible sector (HCSM-15). This paper lifts both results to the full 56-dimensional Hodge sector $V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ at $\lambda = 4$. The lift is implemented by an explicit block diagonal extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ that is a Lie algebra homomorphism, and that implements the natural decomposition of V_{56} into $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with the $\mathbb{C}^2$ vector index of Ω^1 as an inner tensor factor. We prove: (i) the 56-dimensional loop commutant $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}^{(56)})$ is exactly the block-diagonal extension of the 14-dimensional loop commutant, hence a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$; (ii) the hidden sector $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ is a spectator: it does not enlarge, rotate, or otherwise alter the commutant; (iii) the 56-dimensional DNLS commutant is exactly the block-diagonal extension of the 14-dimensional DNLS commutant, hence a 10-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$ isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$; (iv) the full $\mathfrak{su}(2)_L$ factor is preserved exactly on the 56-dimensional sector via the doubled-momentum pairing mechanism of HCSM-15; (v) the isotypic projector degeneracy and the absence of root generators extend to the 56-dimensional sector; (vi) conditional on the identification of the generation index with a 3-dimensional space, the trivial extension $M_{\text{loop}} \otimes \mathbb{I}_3$ and $H_{\text{DNLS}} \otimes \mathbb{I}_3$ to $\mathbb{C}^{168} = \mathbb{C}^{56} \otimes \mathbb{C}^3$ commute with any orthogonal 3×3 action on the generation index, in particular with the unique three dimensional irreducible representation of the alternating group A_4 . All 56-dimensional results are HCSM-native and verified at machine precision. The A_4 identification with the physical generation index is a labeling input, not a derivation.

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1 Introduction

1.1 The 56-dimensional Hodge sector

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04 [4]). On the full Hodge complex at $\lambda = 4$, the 14-mode multiplet lifts to

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad \dim V_{56} = 14 + 28 + 14 = 56 \quad (1)$$

(HCSM-02 [3]). The D_4 isotypic decomposition of V_{56} is

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (2)$$

This 56-dimensional sector is the natural arena for the framework's dynamical operators.

On the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$, HCSM-15 [9] determined the intersection of $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ with the commutant of each of two operators: the loop operator $M_{\text{loop}} = M_A P_A + M_B P_B$ and the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$. The two commutants are respectively 3-dimensional (Hermitian, abelian) and 10-dimensional (isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$).

The present paper lifts both HCSM-15 results to the full 56-dimensional Hodge sector.

1.2 The one line summary

The 56-dimensional lift of the gauge algebra, the loop operator, and the DNLS Hartree Hamiltonian is achieved by an explicit block-diagonal extension map that is a Lie algebra homomorphism. Both commutants on V_{56} are exactly the block-diagonal extensions of their 14-dimensional counterparts. The hidden sector is a spectator: it does not enlarge, rotate, or alter the preserved gauge algebra. Conditional on a 3-dimensional generation index, the framework's operators commute with any orthogonal action on that index, including the A_4 three-dimensional irrep.

1.3 Main results

- **Extension map (Section 3).** The block-diagonal extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ is a Lie algebra homomorphism: $\text{extend}([A, B]) = [\text{extend}(A), \text{extend}(B)]$.
- **Coefficient-space isomorphism (Theorem 8).** The nullspace of the 14-dimensional loop commutator matrix and the nullspace of the 56-dimensional loop commutator matrix are identical subspaces of $\mathbb{R}^{16}$.
- **56-dim loop commutant (Theorem 10).** $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}^{(56)})$ is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$, equal to the block-diagonal extension of the 14-dimensional loop commutant.
- **Hidden-sector neutrality (Theorem 11).** The hidden sector does not enlarge, rotate, or otherwise alter the loop commutant.
- **56-dim DNLS commutant (Theorem 15).** $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}^{(56)})$ is a 10-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$ isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$, equal to the block-diagonal extension of the 14-dimensional DNLS commutant.
- **$\mathfrak{su}(2)_L$ preservation on 56-dim (Theorem 16).** The full $\mathfrak{su}(2)_L$ factor commutes with $H_{\text{DNLS}}^{(56)}$ exactly.

- **Isotypic projector degeneracy and no-root (Theorems 19 and 20).** Both structural features of HCSM-15 extend to the 56-dimensional sector.
- **Conditional generation extension (Theorem 22).** Let n be the generation count. The trivial extensions $M_{\text{loop}} \otimes \mathbb{I}_n$, $H_{\text{DNLS}} \otimes \mathbb{I}_n$ to $\mathbb{C}^{56n}$ commute with $\mathbb{I}_{56} \otimes B$ for any $B \in M_n(\mathbb{C})$, in particular for the 3-dimensional orthogonal action of A_4 when $n = 3$.

1.4 Cross references

The paper depends on the following HCSM companions:

- HCSM-00 [1]: foundations, notation, mapping.
- HCSM-01 [2]: the 8×8 torus substrate.
- HCSM-02 [3]: the Hodge complex on Λ ; the D_4 decomposition of V_{56} .
- HCSM-04 [4]: the 14-mode multiplet at $\lambda = 4$.
- HCSM-05 [5]: orbit decomposition and inverse participation.
- HCSM-06 [6]: the gauge algebra $\mathfrak{g}_{\text{SM}}$.
- HCSM-13 [7]: the processor T ; the F/B -class correspondence.
- HCSM-14 [8]: the winding generator spectrum and class partition.
- HCSM-15 [9]: gauge symmetry of the loop operator and DNLS on the 14-dim visible sector.
- HCSM-17 [10]: Cartan rotation angles on the loop commutant.

1.5 Organization

Section 2 states the 56-dimensional Hodge sector and its D_4 structure. Section 3 defines the extension map and proves it is a Lie algebra homomorphism. Section 4 proves the 56-dimensional loop commutant theorem and the hidden-sector neutrality theorem. Section 5 proves the 56-dimensional DNLS commutant theorem and the exact $\mathfrak{su}(2)_L$ preservation. Section 6 extends the isotypic projector degeneracy and the no-root results. Section 7 states and proves the conditional generation extension to 168 dimensions. Section 8 catalogs inputs and derived. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup: the 56-dimensional Hodge sector

2.1 Substrate and Hodge complex

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The Hodge complex of differential forms is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad \dim = 256, \quad (3)$$

and the Hodge-Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree [3]. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (4)$$

At the self-paired eigenvalue $\lambda = 4$, the mode set has 14 elements (HCSM-04 [4]).

2.2 The 56-dimensional sector at $\lambda = 4$

Restricting the Hodge complex to the $\lambda = 4$ eigenvalue gives the 56-dimensional sector

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (5)$$

with dimensions $14 + 28 + 14 = 56$. Under D_4 (HCSM-02 [3], HCSM-05 [5]),

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (6)$$

2.3 Tensor factor ordering

We fix once and for all the tensor factor ordering

$$\mathbb{C}^{56} = \underbrace{\mathbb{C}^{14}}_{\Omega^0} \oplus \underbrace{\mathbb{C}^{14} \otimes \mathbb{C}^2}_{\Omega^1} \oplus \underbrace{\mathbb{C}^{14}}_{\Omega^2}. \quad (7)$$

Indices 0 through 13 span Ω^0 ; indices 14 through 41 span Ω^1 with the $\mathbb{C}^2$ vector index as inner tensor factor ordered as $(14 + 2i + \mu)$ for mode $i \in \{0, \dots, 13\}$ and vector component $\mu \in \{0, 1\}$; indices 42 through 55 span Ω^2 .

2.4 The D_4 action on V_{56}

In the factorized representation

$$\Omega^k|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes \Lambda^k(E), \quad (8)$$

where E is the two-dimensional vector irrep of D_4 and $\Lambda^k(E)$ is its k -th exterior power, the D_4 action on V_{56} is block-diagonal. The generators $R = r|_{\lambda=4}$ and $S = s|_{\lambda=4}$ satisfy

$$R^4 = \mathbb{I}, \quad S^2 = \mathbb{I}, \quad (RS)^2 = \mathbb{I} \quad (9)$$

at machine precision on V_{56} .

2.5 Isotypic projectors on V_{56}

The isotypic projectors on V_{56} are

$$P_\rho^{(56)} = \frac{\dim \rho}{|D_4|} \sum_{g \in D_4} \chi_\rho(g)^* R(g), \quad (10)$$

where $R(g)$ is the representation of g on V_{56} and χ_ρ is the character of the irrep ρ . Direct computation gives

$$\text{tr } P_{A_1} = 9, \quad \text{tr } P_{A_2} = 5, \quad \text{tr } P_{B_1} = 7, \quad \text{tr } P_{B_2} = 7, \quad \text{tr } P_E = 28, \quad (11)$$

each projector idempotent to machine precision, and $\sum_\rho P_\rho^{(56)} = \mathbb{I}_{56}$ exactly.

2.6 Gauge algebra on V_{56}

By HCSM-06 [6], the gauge algebra

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5 \quad (12)$$

is a Lie subalgebra of the D_4 commutant on the 14-mode multiplet. The 16 gauge generators on $\mathbb{C}^{14}$ are

$$\mathfrak{g}_{\text{SM}}^{(14)} = \{T_c^1, \dots, T_c^8, T_L^1, T_L^2, T_L^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}, \quad (13)$$

with $T_c^a = U_{A_1} \lambda^a U_{A_1}^T$ and $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ for orthonormal isotypic bases $U_{A_1} \in \mathbb{R}^{14 \times 3}$, $U_{B_1} \in \mathbb{R}^{14 \times 2}$.

3 The extension map and its properties

3.1 Definition of the extension

Definition 1 (Extension map). The extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ is defined by

$$\text{extend}(G) = \begin{pmatrix} G & 0 & 0 \\ 0 & G \otimes \mathbb{I}_2 & 0 \\ 0 & 0 & G \end{pmatrix} \quad (14)$$

in the block ordering (7).

The extension map is $\mathbb{C}$ -linear and injective. It acts on each Hodge form degree by tensoring the 14-dimensional operator with the appropriate trivial factor: identity on Ω^0 , $G \otimes \mathbb{I}_2$ on Ω^1 , and identity on Ω^2 .

3.2 Lie algebra homomorphism

Lemma 2 (Extension is a Lie algebra homomorphism). *For all $A, B \in \mathbb{C}^{14 \times 14}$,*

$$[\text{extend}(A), \text{extend}(B)] = \text{extend}([A, B]). \quad (15)$$

Proof. Write $\text{extend}(A) = A \oplus (A \otimes \mathbb{I}_2) \oplus A$ in block-diagonal form. For two block-diagonal matrices $M = M_0 \oplus M_1 \oplus M_2$ and $N = N_0 \oplus N_1 \oplus N_2$, the commutator is block-diagonal:

$$[M, N] = [M_0, N_0] \oplus [M_1, N_1] \oplus [M_2, N_2]. \quad (16)$$

Substituting $M = \text{extend}(A)$, $N = \text{extend}(B)$:

$$[\text{extend}(A), \text{extend}(B)] = [A, B] \oplus ([A, B] \otimes \mathbb{I}_2) \oplus [A, B] = \text{extend}([A, B]). \quad (17)$$

□

Corollary 3. *extend restricts to a Lie algebra homomorphism on any Lie subalgebra of $\text{Mat}_{14}(\mathbb{C})$.*

3.3 The 56-dimensional gauge generators

Definition 4 (Gauge generators on V_{56}). The 56-dimensional gauge generators are

$$\mathfrak{g}_{\text{SM}}^{(56)} = \{\text{extend}(g_1^{(14)}), \dots, \text{extend}(g_{16}^{(14)})\}. \quad (18)$$

Each $\text{extend}(g_j^{(14)})$ is Hermitian and D_4 -invariant on V_{56} .

Remark 5. The Hermiticity is immediate: $\text{extend}(G)^\dagger = \text{extend}(G^\dagger) = \text{extend}(G)$ for Hermitian G . The D_4 -invariance follows from the block-diagonal structure (14) and the fact that the D_4 action respects the form-degree decomposition $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

4 The loop commutant on V_{56}

4.1 The loop operator on V_{56}

Definition 6 (Loop operator on V_{56}). The 56-dimensional loop operator is

$$M_{\text{loop}}^{(56)} = \text{extend}(M_{\text{loop}}^{(14)}), \quad M_{\text{loop}}^{(14)} = M_A P_A^{(14)} + M_B P_B^{(14)}, \quad (19)$$

where M_A, M_B are the complex loop eigenvalues of HCSM-14 [8] and HCSM-17 [10], and $P_A^{(14)}, P_B^{(14)}$ are the class projectors of HCSM-14 [8].

$M_{\text{loop}}^{(56)}$ is block-diagonal, non-Hermitian (dissipative), and satisfies

$$[M_{\text{loop}}^{(56)}, P_A^{(56)}] = [M_{\text{loop}}^{(56)}, P_B^{(56)}] = [M_{\text{loop}}^{(56)}, P_C^{(56)}] = 0, \quad (20)$$

where $P_A^{(56)}, P_B^{(56)}, P_C^{(56)}$ are the 56-dimensional class projectors obtained by extending the 14-dimensional class projectors.

4.2 Coefficient-space isomorphism

Definition 7 (Commutator matrices). For $d \in \{14, 56\}$, define the commutator matrix $C^{(d)} \in \mathbb{C}^{(d)^2 \times 16}$ by

$$C_{k,j}^{(d)} = [M_{\text{loop}}^{(d)}, g_j^{(d)}]_k, \quad k = 1, \dots, d^2, \quad j = 1, \dots, 16. \quad (21)$$

Let $N^{(d)} = \{v \in \mathbb{R}^{16} : C_{\mathbb{R}}^{(d)} v = 0\}$ be the real nullspace, where $C_{\mathbb{R}}^{(d)} = [\text{Re}(C^{(d)}); \text{Im}(C^{(d)})]$.

Theorem 8 (Coefficient-space identity). *The nullspaces $N^{(14)}$ and $N^{(56)}$ are identical subspaces of $\mathbb{R}^{16}$.*

Proof. Let $v \in \mathbb{R}^{16}$. By Lemma 2 and linearity of extend,

$$\sum_{j=1}^{16} v_j [M_{\text{loop}}^{(56)}, g_j^{(56)}] = \left[M_{\text{loop}}^{(56)}, \sum_j v_j g_j^{(56)} \right] = \left[\text{extend}(M_{\text{loop}}^{(14)}), \text{extend} \left(\sum_j v_j g_j^{(14)} \right) \right] = \text{extend} \left(\left[M_{\text{loop}}^{(14)}, \sum_j v_j g_j^{(14)} \right] \right) \quad (22)$$

The extension map is injective, so this vanishes iff $[M_{\text{loop}}^{(14)}, \sum_j v_j g_j^{(14)}] = 0$. Hence $v \in N^{(56)} \Leftrightarrow v \in N^{(14)}$. $\square$

Corollary 9 (Explicit isomorphism of commutants). *The map*

$$\Phi : \text{Comm}^{(14)}(M_{\text{loop}}) \rightarrow \text{Comm}^{(56)}(M_{\text{loop}}), \quad \sum_j v_j g_j^{(14)} \mapsto \sum_j v_j \text{extend}(g_j^{(14)}) \quad (23)$$

is a linear isomorphism of real vector spaces. On coefficient space, Φ acts as the identity on the common nullspace $N \subseteq \mathbb{R}^{16}$.

4.3 The 56-dimensional loop commutant theorem

Theorem 10 (Loop commutant dimension on V_{56}). *The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}^{(56)})$ is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$. It is exactly the block-diagonal extension of the 14-dimensional loop commutant:*

$$\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}^{(56)}) = \text{extend}\left(\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}^{(14)})\right). \quad (24)$$

Proof. Dimension. Form the 56-dimensional commutator matrix $C^{(56)} \in \mathbb{C}^{56^2 \times 16}$. The realified matrix $C_{\mathbb{R}}^{(56)} \in \mathbb{R}^{2 \cdot 56^2 \times 16}$ has rank 13 at machine precision, with a singular value gap of $\sim 10^{15}$ between the 13th and 14th singular values. Nullity = $16 - 13 = 3$.

Explicit isomorphism. By Theorem 8, $N^{(14)} = N^{(56)}$. Choosing a common nullspace basis $\{v^{(1)}, v^{(2)}, v^{(3)}\}$, the 56-dimensional commutant has basis

$$O_i^{(56)} = \sum_j v_j^{(i)} \text{extend}(g_j^{(14)}) = \text{extend}(O_i^{(14)}), \quad i = 1, 2, 3. \quad (25)$$

Hence the 56-dimensional commutant equals extend of the 14-dimensional commutant.

Hermiticity. Each $O_i^{(56)}$ is a real linear combination of the Hermitian generators $\text{extend}(g_j^{(14)})$, so it is Hermitian.

Abelian. Cross-commutators vanish by Lemma 2:

$$[O_i^{(56)}, O_j^{(56)}] = [\text{extend}(O_i^{(14)}), \text{extend}(O_j^{(14)})] = \text{extend}([O_i^{(14)}, O_j^{(14)}]) = \text{extend}(0) = 0. \quad (26)$$

□

Theorem 11 (Hidden-sector neutrality). *The hidden sector $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ does not enlarge, rotate, or otherwise alter the loop commutant.*

Proof. Immediate from Theorem 10: the nullspace dimension is 3 on both the 14- and 56-dimensional sectors, and the nullspace coefficient vectors are identical (Theorem 8). The block-diagonal structure (14) of the extension map shows that no mixing occurs between Ω^0 , Ω^1 , and Ω^2 ; the $\mathbb{C}^2$ vector index on Ω^1 is a spectator. □

Remark 12. Theorem 11 is the structural reason the framework's gauge content is independent of the form degree. The loop's symmetry content is determined by the winding class partition (HCSM-14 [8]), not by the form-degree structure of the Hodge complex.

5 The DNLS commutant on V_{56}

5.1 The DNLS Hartree Hamiltonian on V_{56}

Definition 13 (Hartree kernel on V_{56}). Let $K^{(14)} \in \mathbb{R}^{14 \times 14}$ be the Hartree kernel on the 14-dimensional visible sector:

$$K_{ab}^{(14)} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad a, b = 1, \dots, 14, \quad (27)$$

where ϕ_a are the normalized mode wavefunctions on the torus (HCSM-04 [4]). The 56-dimensional extension is

$$K^{(56)} = \text{extend}(K^{(14)}). \quad (28)$$

Definition 14 (DNLS Hartree Hamiltonian on V_{56}). The 56-dimensional DNLS Hartree Hamiltonian is

$$H_{\text{DNLS}}^{(56)} = 4 \mathbb{I}_{56} + UK^{(56)}, \quad U = B_{\text{mass}} = 0.541197. \quad (29)$$

The block-diagonal structure (14) of $K^{(56)}$ ensures that $H_{\text{DNLS}}^{(56)} = \text{extend}(H_{\text{DNLS}}^{(14)})$.

5.2 The 56-dimensional DNLS commutant theorem

Theorem 15 (DNLS commutant dimension on V_{56}). *The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}^{(56)})$ is a 10-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$ isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$. It is exactly the block-diagonal extension of the 14-dimensional DNLS commutant:*

$$\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}^{(56)}) = \text{extend}\left(\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}^{(14)})\right). \quad (30)$$

Proof. Since $H_{\text{DNLS}}^{(56)} = \text{extend}(H_{\text{DNLS}}^{(14)})$ and Lemma 2 applies to $[H_{\text{DNLS}}^{(56)}, g_j^{(56)}]$, the same coefficient-space identity argument as Theorem 8 gives

$$N_{\text{DNLS}}^{(56)} = N_{\text{DNLS}}^{(14)} \subseteq \mathbb{R}^{16}. \quad (31)$$

The dimension is 10 on both sectors, as proved in HCSM-15 [9] (Theorem 5.4). The isomorphism is explicit: $O_i^{(56)} = \text{extend}(O_i^{(14)})$ for each nullspace basis vector $v^{(i)}$.

The algebraic structure (isomorphism to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$) is preserved by the extension map, since extend is a Lie algebra homomorphism (Lemma 2). The bracket relations among the ten generators coincide with the 14-dimensional case. $\square$

5.3 Exact $\mathfrak{su}(2)_L$ preservation on V_{56}

Theorem 16 ($\mathfrak{su}(2)_L$ preservation on V_{56}). *The full $\mathfrak{su}(2)_L$ factor commutes with $H_{\text{DNLS}}^{(56)}$ exactly:*

$$[\text{extend}(T_L^i), H_{\text{DNLS}}^{(56)}] = 0, \quad i = 1, 2, 3. \quad (32)$$

The overlap of each $\text{extend}(T_L^i)$ with the DNLS nullspace is 1.000 at machine precision.

Proof. By HCSM-15 [9] Theorem 6.3, $[T_L^i, H_{\text{DNLS}}^{(14)}] = 0$ for each i . Applying Lemma 2:

$$[\text{extend}(T_L^i), \text{extend}(H_{\text{DNLS}}^{(14)})] = \text{extend}([T_L^i, H_{\text{DNLS}}^{(14)}]) = \text{extend}(0) = 0. \quad (33)$$

The overlap is inherited directly from the 14-dimensional result. $\square$

Remark 17. The doubled-momentum pairing mechanism of HCSM-15 [9] (Theorem 6.1) operates entirely within the B_1 isotypic sector of $\Omega^0|_{\lambda=4}$. On V_{56} , the extension map preserves this mechanism on each form degree: on Ω^0 the mechanism operates directly, on Ω^1 it operates on the $\mathbb{C}^{14}$ factor with the $\mathbb{C}^2$ vector index as a spectator, and on Ω^2 it operates directly. The exact cancellation $K_{B_1}^{(56)} = 0$ follows.

Remark 18 ($\mathfrak{su}(3)_c$ breaking on V_{56}). By the same extension argument, the $\mathfrak{su}(3)_c$ breaking pattern of HCSM-15 [9] Theorem 7.1 lifts to V_{56} : the root generators $\text{extend}(\lambda^2), \text{extend}(\lambda^5), \text{extend}(\lambda^7)$ are orthogonal to the 56-dimensional DNLS commutant, and the 2-dimensional abelian subalgebra $\text{extend}(\mathfrak{c})$ is preserved.

6 Structural results on V_{56}

6.1 Isotypic projector degeneracy

Theorem 19 (Isotypic projector degeneracy on V_{56}). *Every element of the 56-dimensional loop commutant has equal coefficients on the isotypic projectors $P_{A_2}^{(56)}, P_{B_2}^{(56)}, P_E^{(56)}$.*

Proof. By Theorem 10, every element of the 56-dimensional loop commutant is $\text{extend}(O^{(14)})$ for some $O^{(14)}$ in the 14-dimensional loop commutant. By HCSM-15 [9] Theorem 4.1, the coefficient vector of $O^{(14)}$ satisfies $|v_{A_2} - v_{B_2}| < 10^{-15}$, $|v_{A_2} - v_E| < 10^{-15}$. Since the coefficient vector of $O^{(56)}$ is the same v (Theorem 8), the degeneracy is inherited. $\square$

6.2 Absence of root generators

Theorem 20 (No root generators on V_{56}). *The 56-dimensional loop commutant has zero overlap with $\text{extend}(\lambda^a)$ for $a \in \{1, 2, 4, 5, 6, 7\}$ and $\text{extend}(\sigma^i)$ for $i \in \{1, 2\}$. Only the Cartan generators $\text{extend}(\lambda^3), \text{extend}(\lambda^8), \text{extend}(\sigma^3)$ of $\mathfrak{g}_{\text{SM}}$ have nonzero overlap with the 56-dimensional loop commutant.*

Proof. By HCSM-15 [9] Theorem 4.3, the corresponding 14-dimensional coefficient vectors have zero projection onto the 14-dimensional nullspace. By Theorem 8, the 56-dimensional coefficient vectors are identical, so the projection is also zero. $\square$

7 Conditional generation extension to 168 dimensions

7.1 Trivial extension to n generations

Definition 21 (Trivial generation extension). Let n be a positive integer. The trivial generation extension of a linear operator A on $\mathbb{C}^{56}$ is

$$A^{(n)} = A \otimes \mathbb{I}_n \in \mathbb{C}^{56n \times 56n}. \quad (34)$$

Theorem 22 (Trivial extension commutes with generation action). *For any operator $A \in \mathbb{C}^{56 \times 56}$ and any matrix $B \in \mathbb{C}^{n \times n}$,*

$$[A \otimes \mathbb{I}_n, \mathbb{I}_{56} \otimes B] = 0. \quad (35)$$

In particular, the trivially-extended loop operator $M_{\text{loop}}^{(56)} \otimes \mathbb{I}_n$ and DNLS Hamiltonian $H_{\text{DNLS}}^{(56)} \otimes \mathbb{I}_n$ commute with any orthogonal action of a finite group on the n -dimensional generation index.

Proof. The tensor product identity $(A \otimes \mathbb{I}_n)(\mathbb{I}_{56} \otimes B) = A \otimes B = (\mathbb{I}_{56} \otimes B)(A \otimes \mathbb{I}_n)$ gives the commutator directly. Orthogonality of B is not required for commutation; any $B \in M_n(\mathbb{C})$ commutes with $A \otimes \mathbb{I}_n$. $\square$

7.2 The A_4 three-dimensional representation

Definition 23 (A_4 three-dimensional irrep). The alternating group A_4 has 12 elements and a unique three-dimensional irreducible representation $\rho_3 : A_4 \rightarrow SO(3)$. The character of ρ_3 is

$$\chi_{\rho_3}(e) = 3, \quad \chi_{\rho_3}(\text{double transposition}) = -1, \quad \chi_{\rho_3}(\text{3-cycle}) = 0. \quad (36)$$

A concrete realization is the rotation group of the regular tetrahedron acting on the ambient $\mathbb{R}^3$.

Corollary 24 (Conditional generation-extension theorem). *Conditional on the identification of the generation index with a 3-dimensional space, the trivially-extended operators $M_{\text{loop}}^{(56)} \otimes \mathbb{I}_3$ and $H_{\text{DNLS}}^{(56)} \otimes \mathbb{I}_3$ on $\mathbb{C}^{168} = \mathbb{C}^{56} \otimes \mathbb{C}^3$ commute with $\mathbb{I}_{56} \otimes \rho_3(a)$ for every $a \in A_4$:*

$$[M_{\text{loop}}^{(56)} \otimes \mathbb{I}_3, \mathbb{I}_{56} \otimes \rho_3(a)] = 0, \quad [H_{\text{DNLS}}^{(56)} \otimes \mathbb{I}_3, \mathbb{I}_{56} \otimes \rho_3(a)] = 0. \quad (37)$$

The 168-dimensional commutant of the trivially-extended loop with the combined algebra $\mathfrak{g}_{\text{SM}} \oplus A_4$ is 15-dimensional, decomposing as $3 \oplus 12$ with zero overlap.

Proof. Apply Theorem 22 with $n = 3$ and $B = \rho_3(a)$.

The dimension count: the 56-dimensional loop commutant restricted to $\mathfrak{g}_{\text{SM}}$ is 3-dimensional; the A_4 generators are 12 linearly independent elements of $\mathbb{I}_{56} \otimes M_3(\mathbb{C})$ spanning a 12-dimensional space; the two subspaces act on orthogonal tensor factors and are orthogonal under the Frobenius inner product. Direct computation gives total nullity 15 with the stated decomposition. $\square$

Remark 25 (Status of the A_4 identification). The identification of the 3-dimensional A_4 irrep with the physical generation index is a physical input. The framework's derivation of the generation count $d - 1 = 3$ is the subject of HCSM-31. Theorem 22 is pure linear algebra and holds for any n ; the special value $n = 3$ enters only through the identification. This labeling convention is the same kind the Standard Model makes when it labels the three fermion families as "generations."

8 Inputs and derived

8.1 Derived

The following results are proved in this paper:

1. The extension map is a Lie algebra homomorphism (Lemma 2).
2. The 14-dimensional and 56-dimensional loop commutator matrices have identical nullspaces (Theorem 8).
3. The 56-dimensional loop commutant is 3-dimensional, real, Hermitian, abelian, and equal to the block-diagonal extension of the 14-dimensional loop commutant (Theorem 10).
4. Hidden-sector neutrality: the hidden sector does not enlarge, rotate, or alter the loop commutant (Theorem 11).
5. The 56-dimensional DNLS commutant is 10-dimensional, real, Hermitian, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$, and equal to the block-diagonal extension of the 14-dimensional DNLS commutant (Theorem 15).
6. Exact $\mathfrak{su}(2)_L$ preservation on V_{56} (Theorem 16).
7. Isotypic projector degeneracy on V_{56} (Theorem 19).
8. No root generators on V_{56} (Theorem 20).
9. Trivial generation extension commutes with any $n \times n$ action on the generation index (Theorem 22).
10. Conditional A_4 generation-extension theorem (Corollary 24).

8.2 Inputs

1. The 56-dimensional Hodge sector $V_{56} = \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at $\lambda = 4$ and its D_4 decomposition (HCSM-02).
2. The 14-mode multiplet and its mode wavefunctions (HCSM-04).
3. The gauge algebra $\mathfrak{g}_{\text{SM}}$ and the isotypic embeddings (HCSM-06).
4. The processor T and the class partition (HCSM-13).
5. The winding generator and class partition (HCSM-14).
6. The 14-dimensional loop commutant and DNLS commutant results (HCSM-15).
7. The complex loop eigenvalues M_A, M_B and their ratio (HCSM-14, HCSM-17).
8. The mass-map rate $U = B_{\text{mass}} = 0.541197$ (HCSM-33).
9. The 3-dimensional orthogonal representation of A_4 (standard finite group theory).
10. The identification of the 3-dimensional generation index with the physical generation space (physical input, consistent with the framework’s labeling convention; the derivation of the generation count is the subject of HCSM-31).

8.3 What is not claimed

We do not claim to derive the gauge algebra, the class partition, or the loop eigenvalues from the substrate; those are the subjects of HCSM-06, HCSM-14, and HCSM-17. We do not claim to derive the specific form of the A_4 action from the substrate; the identification of the generation index with the 3-dimensional A_4 irrep is a physical input, and the derivation of the generation count $d - 1 = 3$ is the subject of HCSM-31. We do not claim to compute the Cartan rotation angles on the 56-dimensional sector; those are the subject of HCSM-17.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at 12-VERIFICATION/HCSM-16_verification.py. It constructs:

1. The 56-dimensional Hodge sector at $\lambda = 4$ and its D_4 action.
2. The isotypic projectors $P_{A_1}^{(56)}, \dots, P_E^{(56)}$.
3. The 56-dimensional gauge generators by the extension map (14).
4. The 56-dimensional loop operator $M_{\text{loop}}^{(56)}$.
5. The 56-dimensional DNLS Hartree kernel $K^{(56)}$ and Hamiltonian $H_{\text{DNLS}}^{(56)}$.
6. The 168-dimensional trivially extended operators.

and verifies:

1. extend is a Lie algebra homomorphism.
2. The 56-dimensional loop commutant rank 13, nullity 3; the nullspace equals the 14-dimensional nullspace (Theorem 8).
3. Hidden-sector neutrality: the 56-dimensional nullspace dimension is 3 (Theorem 11).
4. The 56-dimensional DNLS commutant rank 6, nullity 10.
5. $\mathfrak{su}(2)_L$ overlap 1 for $\text{extend}(\sigma^1), \text{extend}(\sigma^2), \text{extend}(\sigma^3)$ on V_{56} .
6. Isotypic projector degeneracy on V_{56} .
7. Zero overlap of $\text{extend}(\lambda^{1,2,4,5,6,7})$ and $\text{extend}(\sigma^{1,2})$ with the 56-dimensional loop commutant.
8. The trivial extension commutes with $\mathbb{I}_{56} \otimes B$ for arbitrary $B \in M_3(\mathbb{C})$.
9. The 168-dimensional commutant nullity 15, decomposing as $3 \oplus 12$ with zero overlap.

Running 12-VERIFICATION/run_all.py reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

This paper has extended the loop operator and the DNLS Hartree Hamiltonian from the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$ to the full 56-dimensional Hodge sector $V_{56} = \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at $\lambda = 4$. The extension is implemented by an explicit block-diagonal map that is a Lie algebra homomorphism and that respects the natural $\mathbb{C}^2$ tensor factor on Ω^1 .

The main structural results are:

- **Loop commutant is universal.** The 56-dimensional loop commutant $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}^{(56)})$ is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$, equal to the block-diagonal extension of the 14-dimensional loop commutant. The hidden sector $\Omega^1 \oplus \Omega^2$ is a spectator: it does not enlarge, rotate, or alter the commutant.
- **DNLS commutant is universal.** The 56-dimensional DNLS commutant $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}^{(56)})$ is a 10-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$ isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$, equal to the block-diagonal extension of the 14-dimensional DNLS commutant. The full $\mathfrak{su}(2)_L$ factor commutes with $H_{\text{DNLS}}^{(56)}$ exactly.
- **Structural features extend.** The isotypic projector degeneracy $P_{A_2}^{(56)} = P_{B_2}^{(56)} = P_E^{(56)}$ on every element of the loop commutant, and the absence of root generators $\text{extend}(\lambda^{1,2,4,5,6,7})$, $\text{extend}(\sigma^{1,2})$, both carry over from the 14-dimensional sector.
- **Generation extension is trivial.** The trivially-extended operators on $\mathbb{C}^{56n}$ commute with any orthogonal action on the n -dimensional generation index. Conditional on the identification of the generation index with a 3-dimensional space, the framework's operators commute with the A_4 three-dimensional irrep.

The recurring structural zero—the equality of the 14-dimensional, 56-dimensional, and 168-dimensional loop commutants restricted to $\mathfrak{g}_{\text{SM}}$ —is the fingerprint of the class partition as the fundamental object of the framework. The loop's symmetry content is determined by the winding generator's class structure, not by the form degree or the generation index.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-05: Orbit Decomposition and Inverse Participation*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-06: Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-13: The Processor T and the F/B -Class Correspondence*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-14: The Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-15: Gauge Symmetry of the Loop Operator and DNLS*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-17: Loop Operator Cartan Rotation*, HCSM White Paper Series (2026).
- [11] S. Preschutti, *HCSM-33: Mass Map*, HCSM White Paper Series (2026).
- [12] I. Schur, *Neue Begründung der Theorie der Gruppencharaktere*, Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (1905), 406–432.
- [13] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. (CRC Press, 2003).

A Web Crawler Metadata

A.1 Dublin Core

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A.2 Schema.org JSON-LD

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           neutrality is proved. Conditional on a 3-dimensional generation
           index, the trivially-extended framework operators commute with
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